





POLITECNICO
MILANO 1863



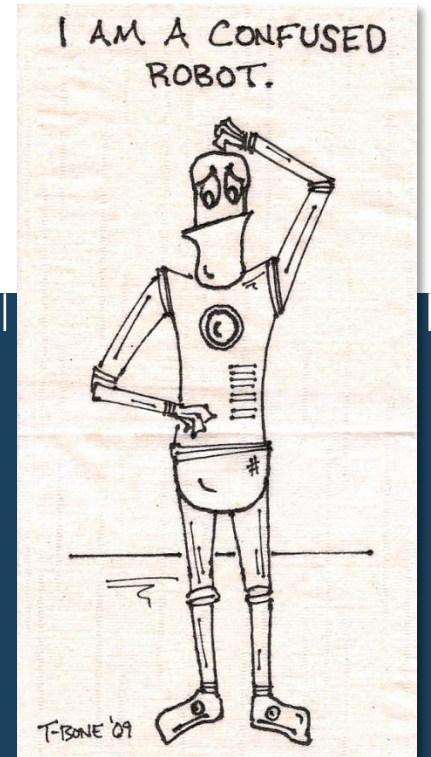
POLITECNICO
MILANO 1863

Robotics

Course Introduction

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Artificial Intelligence and Robotics Lab - Politecnico di Milano



About me and my lectures ...

mobile robotics

Lectures given by Matteo Matteucci

- +39 02 2399 3470
- matteo.matteucci@polimi.it

Research Topics

- Robotics and Autonomous Systems
- Computer Vision and Perception
- Pattern Recognition & Machine Learning
- Benchmarking in Robotics



Aims of these lectures: learning how to design and implement the software which makes autonomous an autonomous mobile robot (e.g., symbolic planning, trajectory planning, localization, perception, mapping, etc.)
↓
autonomy on navigation



... what about the course?

All the infos on the course website

- <https://chrome.deib.polimi.it/index.php?title=Robotics>

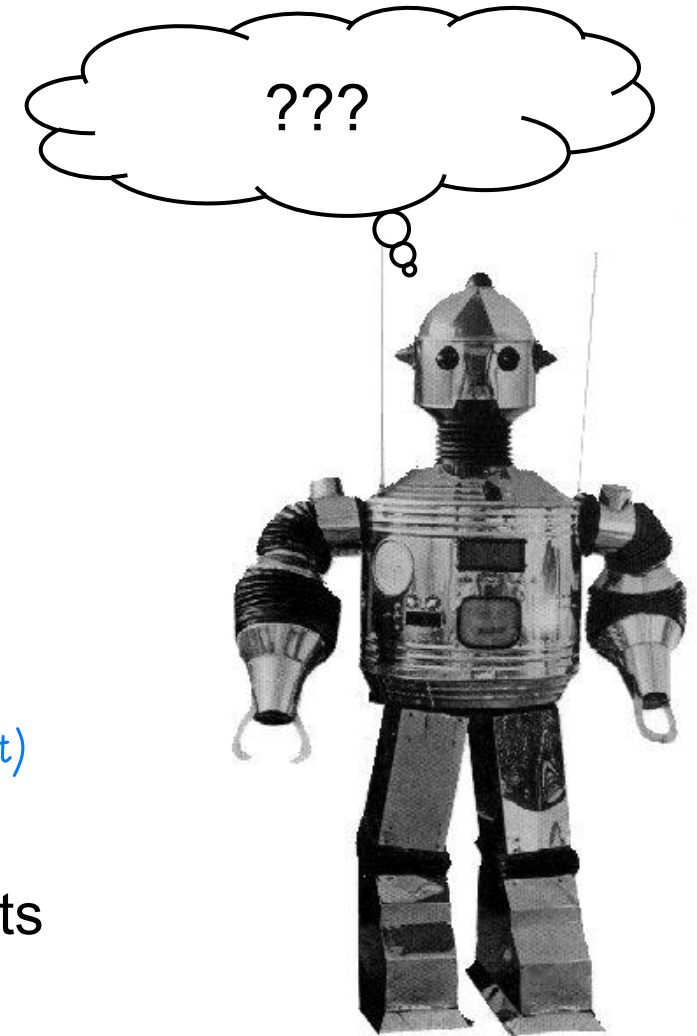
Lectures given by:

- Matteo Matteucci (Lecturer ~30h)
matteo.matteucci@polimi.it
- Simone Mentasti (Teaching Assistant ~20h)
simone.mentasti@polimi.it

Course in code sharing!!

- Robotics
- Perception, localization, and mapping for mobile robots

↳ Lab / exercise
(practical stuff)
ROS (Robot Operating Syst)



Lectures outline / approach

Theory

Introduction to (mobile) robotics

The mind of a mobile robot

Anatomy of a mobile robot

- Sensors and actuators
- Common Kinematics

Localization and Mapping

- Localization vs Mapping
- Simultaneous Localization & Mapping

Robot autonomous navigation

- Motion control and obstacle avoidance
- Trajectory following
- Trajectory planning
(graph and sample based)

«Theory»

by ROS
↑
Lab

like a "library" to write
Robot sw, commonly used
on research Robotics

ROS1
ROS2
↳ for industry
requirements
(also autom.
cars)

Middleware in robotics

- Motivations and state of the art
- ROS Installation party

ROS Basics

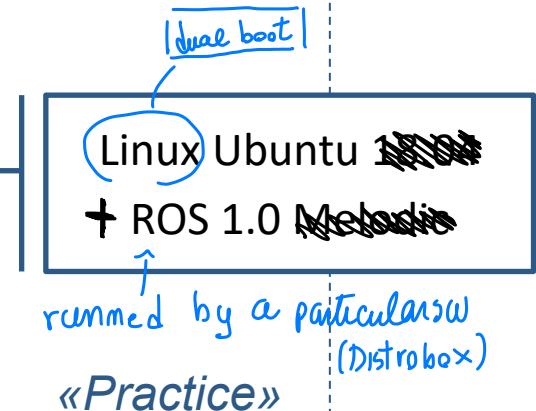
- Publisher/subscriber
- Messages, services, parameters
- Bags, tb, actionlib, rqt_tools
- Message filters, rospy

ROS Advanced

- ROS on multiple machines
- Time synchronization
- Stage

Navigation in ROS

ROS2 & ...



Course organization / rules

Classes (no distinction between lecture and exercise day):

- Wednesday, 12:15 – 14:15, in 26.11
- Thursday, 14:15 – 16:15, in T2.2

Cum tempore!

May change ...

Detailed calendar online (updated weekly)

- <https://chrome.deib.polimi.it/index.php?title=Robotics>

Grading policy:

- Written examination covering the whole program up to
- ROS Home projects graded up to $\rightarrow (2 \text{ ROS projects})$
- Final score will be the sum of the two grades ...

26/32 +

06/32 =

32/32

Deadlines to deliver it,
valid the whole
academic year.

$\downarrow$ problems given as home work $\Rightarrow$ mark valid for 1y than its last!



Alternative projects (instead of exam)

This year we plan to set up 2 teams of students participating to competitions

- Field Robot Event (<https://fieldrobot.nl/event/>) 12th to 15th June 2023
- Leonardo Drone Contest (<https://www.leonardo.com/en/innovation-technology/open-innovation/drone-contest>) October 2023

Real Robots + ROS2 +
physical event + ...

Organization of the teams (4-5 people per team)

- Supervised by a PhD Student / Researcher
- Led by Master Students doing his/her thesis on the topic
- Composed by 3-4 Students from the course

Grading policy:

- The project replace the homeworks
- Written exam replaced by an oral exam with the teacher

I suggest to do the
homeworks anyway!



Course material

Material available on the course website

- Check <https://chrome.deib.polimi.it/index.php?title=Robotics>
- Slides from the teachers (not necessarily available in advance)
- Link to online sources, books and papers
- Link to other websites for tools and digital resources

Past exams and sample questions

- Expect 3 theoretical questions + 2 practical exercises (on average)
- No coding exercise since you have it in the home project
- Few past exam examples are available on the course website

Do you need any further info?

